Pronob Kumar Barman

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EDUCATION

University of Maryland, Baltimore County
PhD in Information Systems, CGPA: 3.85/4.00 — 01/2022 - 05/2025 (Anticipated)
Research Focus: Generative Models, NLP, LLMs, Recommendation Systems

University of Kentucky
MS in Statistics, CGPA: 3.85/4.00 — 08/2019 - 05/2021
Advanced Theoretical Focus: Probability Theory, Statistical Inference, Experimental Design

University of Dhaka
MS in Statistics, CGPA: 3.46/4.00 — 10/2015 - 01/2017
BS in Statistics, CGPA: 3.46/4.00 — 01/2011 - 08/2015

PROFESSIONAL EXPERIENCE

University of Maryland, Baltimore County
Teaching Assistant — 01/2022 – Present

- Designed and conducted workshops on **Machine Learning (ML)**, **LLM workflows**, and system reliability for real-world applications.
- Assisted students in deploying scalable ML pipelines and models on **cloud platforms** like AWS and Vertex AI.

Technuf LLC
Applied Scientist Intern — 06/2024 – 08/2024

- Engineered **AI solutions** by integrating **Large Language Models (LLMs)** such as GPT and BERT with **cybersecurity frameworks (SIEM and SOC systems)**, achieving a **25% improvement in threat detection accuracy**.
- Conducted advanced trend analysis on alert data, reducing **false positives by 20%**, and **synthesized actionable insights** for business impact.

Applied Scientist Intern — 08/2022 – 12/2022

- Streamlined data processing for **200K+ healthcare records** using optimized **SQL queries** in PostgreSQL, improving system efficiency by **40%**.
- Built interactive **Power BI dashboards** to visualize critical healthcare metrics, enabling **data-driven decision-making**.
- Automated testing workflows with **Flutter**, reducing debugging time and improving product release efficiency.

University of Kentucky
Teaching Assistant — 08/2019 – 05/2021

- Delivered hands-on training in **machine learning algorithms**, **data preprocessing techniques**, and **SQL-based data pipelines**, enhancing students' applied problem-solving skills.
- Collaborated on research projects requiring **machine learning model evaluation**, emphasizing performance metrics and ensuring statistical rigor in model validation.

Bangladesh Bank
Deputy Director, Data Science Team — 04/2018 – 12/2021

- Built **statistical models** for credit risk assessment, reducing default rates by **10%**, saving millions in operational costs annually.
- Leveraged **machine learning** and **data mining** to improve **foreign investment strategies**, leading to a **15% increase in FDI inflows**.

SKILLS

- **Programming Languages:** Python, R, Java, Matlab, SAS, SQL
- **ML Frameworks:** TensorFlow, PyTorch, scikit-learn, Keras
- **MLOps Tools:** Docker, Kubernetes
- **Data Science Libraries:** pandas, NumPy, SciPy, NetworkX, StatsModels
- **Generative AI:** GPT-3/4, LangChain, Hugging Face, OpenAI API
- **Visualization Tools:** Tableau, Power BI, Matplotlib, Seaborn, Plotly
- **Natural Language Processing:** BERT, GPT, SpaCy, NLTK
- **Data Management:** PostgreSQL, Pinecone, ChromaDB
- **Cloud Platforms:** AWS sagemaker, Google Cloud, Vertex AI
- **Research Tools:** NVivo, Qualtrics, LaTeX, VScode, Git, Shell (Bash), Jupyter Notebook, R Markdown
- **Soft Skills:** Leadership, Collaboration, Communication, Problem-Solving

PHD RESEARCH PROJECT

Online Healthcare Support Group Formation Using LLMs and NLP Techniques

- Developed two novel generative models: **Group-specific Dirichlet Multinomial Regression (gDMR)** and **Group-specific Structural Topic Model (gSTM)**, aimed at automating and personalizing support group formation in online healthcare platforms.
- The gDMR model extends traditional **Dirichlet Multinomial Regression (DMR)** by introducing group-specific parameters and leveraging user demographics, content, BERT embeddings, and interaction patterns, enabling the formation of semantically coherent and demographically relevant groups.
- The gSTM model enhances **Structural Topic Modeling (STM)** by incorporating group-specific parameters and node embeddings, improving group stability and enabling more personalized recommendations.
- Achieved a **60% improvement in topic coherence** and a **75% boost in recommendation accuracy**, validated through extensive real-world experiments.
- Released open-source implementations, including an R package for gSTM and a Python code for gDMR, facilitating adoption in diverse research and application domains.

SELECTED PROJECTS

DistilBERT-based Question Answering System — GitHub

- Deployed **DistilBERT** models on AWS SageMaker for real-time customer query resolution.
- Reduced response time by **30%** through scalable API integration.

Deploying a Machine Learning Model for Digit Classification on Google Cloud Run — Github

- Developed and deployed a scalable **Flask-based web** app using **TensorFlow**, containerized with **Docker**, and hosted on **Google Cloud Run** to enable real-time predictions through a public API endpoint.

Air Quality Prediction and Deployment on Google Cloud Platform — Github

- Designed and deployed a scalable Air Quality Index (AQI) forecasting system using linear regression, containerized with **Docker**, and integrated with a **Flask** web app on **Google Cloud Platform** for interactive user predictions.

LLM Generative AI with LangChain and OpenAI API — Github

- Developed a **retrieval-augmented generation system** for advanced semantic search, using **Pinecone and ChromaDB**.

End-to-End Medical Chatbot with Llama2 and Hugging Face API — Github

- Developed a **medical chatbot with Llama2**, improving **patient interaction flow by 30%** through **LangChain** enhancements.

Cognitive and Affective State Recognition in iVR Using Multimodal Signals — Github

- Developed inverse inference models using **multimodal signals**, integrating neuro-physiological data such as EEG and eye-tracking to map cognitive and affective states in immersive virtual reality (iVR).

Predictive Policing with GAN

- Implemented a Generative Adversarial Network (GAN) model to identify and mitigate bias in predictive policing algorithms, achieving a 20% reduction in false-positive rates.

PUBLICATIONS

- Balogun, Z. A., **Barman, P. K.**, & Reynolds, T. L. (2025). "Cultivating a Personalized Digital Health Ecosystem: Insights from a Mixed Methods Survey of U.S. Adults." *ACM SIGCHI Conference on Computer-Supported Cooperative Work & Social Computing (CSCW)*, 2025. (Under Review)
- Walid, M. A. A., Devnath, M. K., Muiz, B., Melon, M. M. H., **Barman, P. K.**, & Barman, P. K. (2024). "Efficient Graph-Based Intrusion Detection for CAN Bus Security with Minimal Training Data." *Proceedings of the 6th IEEE International Conference on Sustainable Technologies for Industry 5.0 (STI 2024)*, 2024.
- Balogun, Z. A., **Barman, P. K.**, Onwumbiko, B. K., & Reynolds, T. L. (2024). "User-generated Data for Different Types of Mobile-Personal Health Records: A Computationally-guided Qualitative Analysis Approach." *AMIA Annual Symposium, 2024.*
- Faruq, M. O., Walid, M. A. A., Baowaly, M. K., Devnath, M. K., Ejaz, M. S., **Barman, P. K.**, & Sattar, A. S. (2024). "Shielding Privacy: A Technique of Extenuating Composition Attacks in Various Independent Data Publication." *Bulletin of Electrical Engineering and Informatics, 13(4), 2677-2686.*
- **Barman, P. K.** (2023). "Detecting Abnormal Health Conditions in Smart Home Using a Drone." *arXiv preprint arXiv:2310.05012.*

PEER REVIEWER

- NeurIPS 2024 Workshop on Behavioral ML
- AMIA 2024 Annual Symposium